

Lab 4 – Digital to Analog Conversion

Analog & Digital Communication - EE322

Fall 2026

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Date: _____

Objective

The objective of this lab is to design, compare and contrast the following two techniques of digital to analog signal:

1. Binary weighted resistors

2. R-2R ladder

Introduction

Nowadays digital circuitry is being used everywhere. Connecting the digital circuitry to sensor devices will be easy if the sensor devices are also digital. Electronic components like switches, relays, and encoders are easily interfaced with gate circuits due to the on/off nature of their signals.

In communication system, digital transmission is much faster and convenient, but the digital signals need to be converted back to analog signals at the receiving terminal. In this experiment, we will focus on the digital to analog converter, implementing two different circuit designs.

Binary Weighted Resistors DAC:

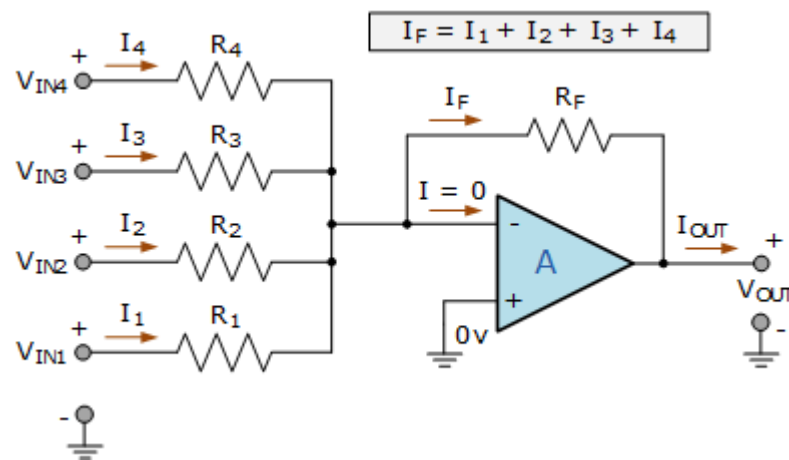


Figure 1: Binary Weighted Resistors DAC. [1]

The binary weighted resistors DAC has n number of switches, one for each bit applied to the input. The resistors of this circuit are inversely proportional to the numerical significance of its binary digit. Equation shows the computation for the output voltage of the binary weighted resistors.

$$V_{OUT} = - \left[\frac{R_F}{R_4} V_D + \frac{R_F}{R_3} V_C + \frac{R_F}{R_2} V_B + \frac{R_F}{R_1} V_A \right] \quad \dots \text{eq 1}$$

R-2R Ladder Network:

R-2R weighted resistor ladder network uses only 2 set of resistors- R and 2R. This is a 4 bit DAC. Let us consider the digital data $D_3D_2D_1D_0 = 0001$ is applied to the DAC, then the Thevenin equivalent circuit reduction is shown on next page.

V_{ref} is nothing but the input binary value voltage, that is for binary 1, $V_{ref}=5V$ and for binary 0, $V_{ref}=0V$.

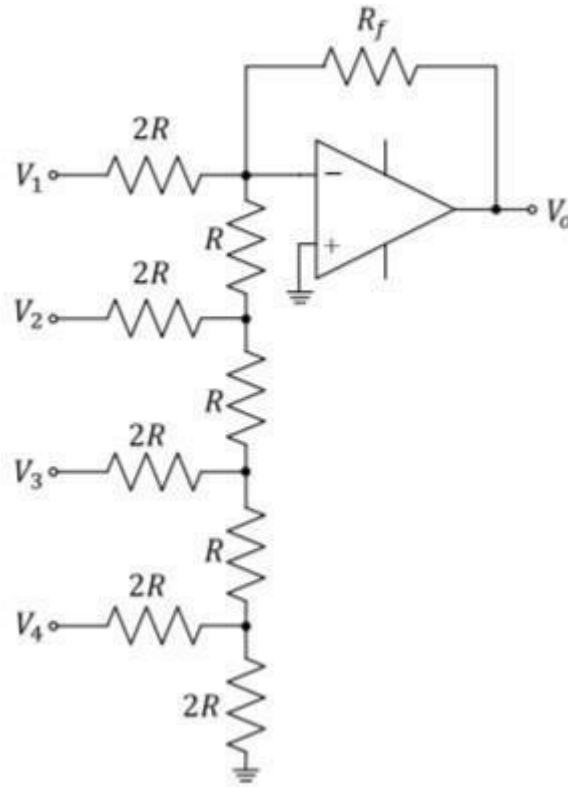


Figure 2: R-2R Ladder Network.

For 0001 only $D_0 = V_{ref}$, all other input are at 0V and can be treated as ground. So finally $V_{ref} / 16$ volt is appearing as the input to op amp. This value get multiplied by the gain of op amp circuit $-(R_f/R_i)$.

If we proceed in this manner (Thevenin equivalent reduction), we will get,

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$$V_{out} = -\frac{R_f}{R_i} \left[\frac{D_0}{16} + \frac{D_1}{8} + \frac{D_2}{4} + \frac{D_3}{2} \right] V_{ref} \quad (2)$$

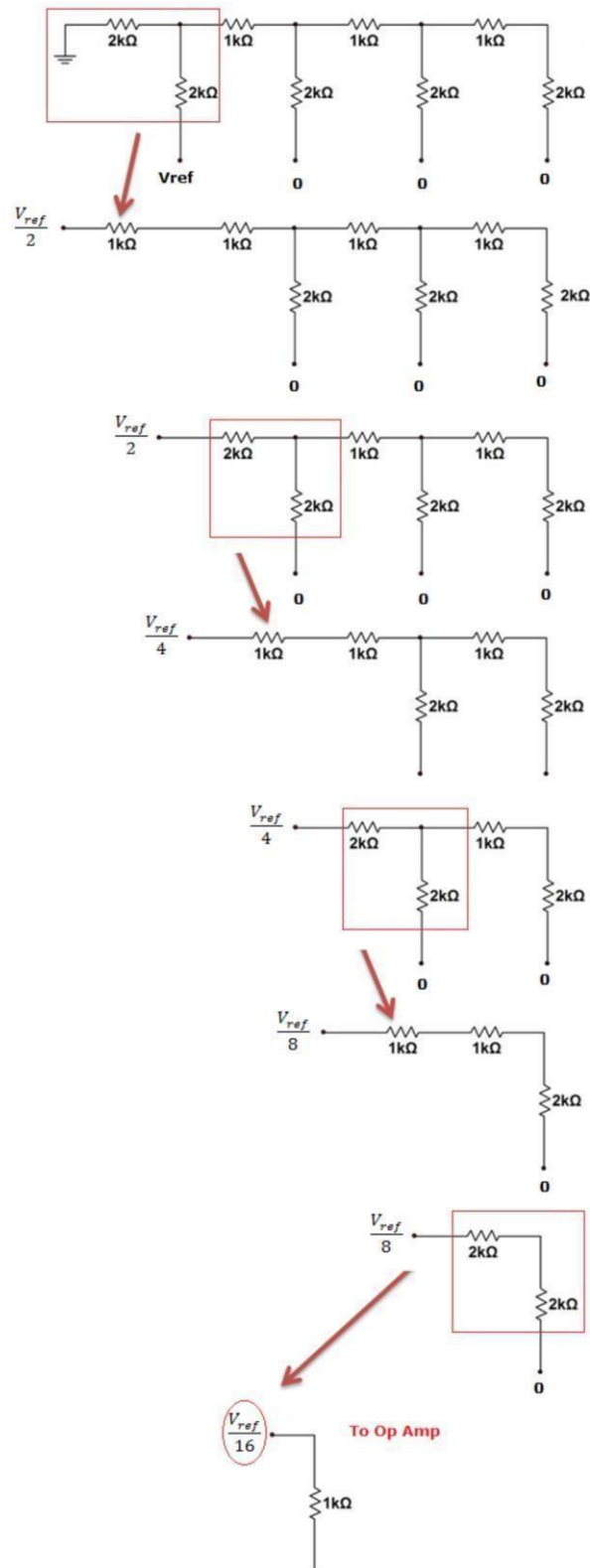


Figure 3: Thevenin equivalent circuit reduction ^[1]

Experimental Work

Apparatus Required

The following apparatus are required to perform this experiment.

No.	Apparatus	Specifications	Quantity
1	DC Power Supply	0-30 V,1 A	1
2	Digital Multi-meter	-	1
3	Breadboard	-	1

Table 1: Apparatus required to perform this experiment.

In-Lab Tasks

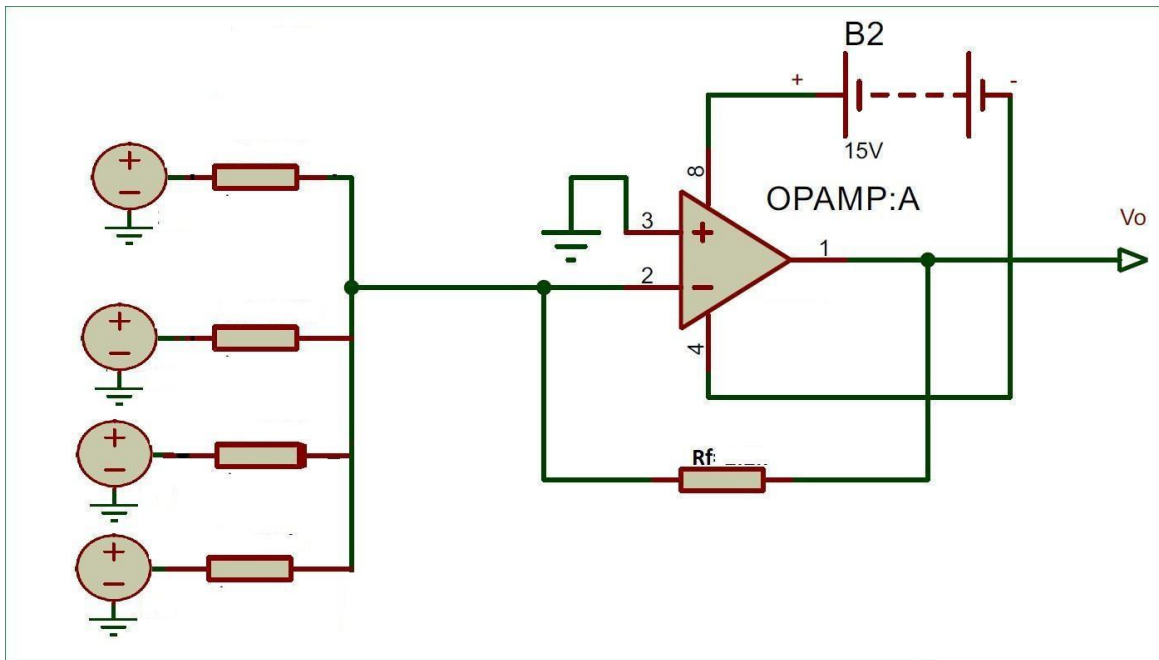
Task 1: Implementing a circuit of binary weighted resistors, DAC.

Components Required

The following components are required to perform this experiment.

No.	Components	Specifications / Values	Quantity
1	Resistors	use different resistors in combination to make binary weighted resistors	---
2	Op-Amp	LM741	1

Procedure



Implement the following circuit shown

1. Using an equation 1. calculate the theoretical values of binary weighted resistors, DAC, output and record them in.
2. Obtain all the experimental values of binary weighted resistors,
3. Measure voltage level using DMM and record them in Table
4. Now calculate percentage error in experimental value by following formula,

$$\text{Percentage Error} = \frac{\text{theoretical value} - \text{experimental value}}{\text{theoretical value}} \times 100$$

S.No	Binary	BWR theory	BWR experiment	Error (%)
0	0 0 0 0	0.000	0.002	N/A
1	0 0 0 1	-0.625	-0.631	0.96
2	0 0 1 0	-1.250	-1.242	0.64
3	0 0 1 1	-1.875	-1.846	1.55
4	0 1 0 0	-2.500	-2.517	0.68
5	0 1 0 1	-3.125	-3.111	0.45
6	0 1 1 0	-3.750	-3.740	0.27
7	0 1 1 1	-4.375	-4.369	0.14
8	1 0 0 0	-5.000	-4.980	0.40

9	1 0 0 1	-5.625	-5.610	0.27
10	1 0 1 0	-6.250	-6.173	1.23
11	1 0 1 1	-6.875	-6.863	0.17
12	1 1 0 0	-7.500	-7.492	0.11
13	1 1 0 1	-8.125	-8.160	0.43
14	1 1 1 0	-8.750	-8.740	0.11
15	1 1 1 1	-9.375	-9.369	0.06

Task 2: Implementing a circuit R-2R ladder network.

Components Required

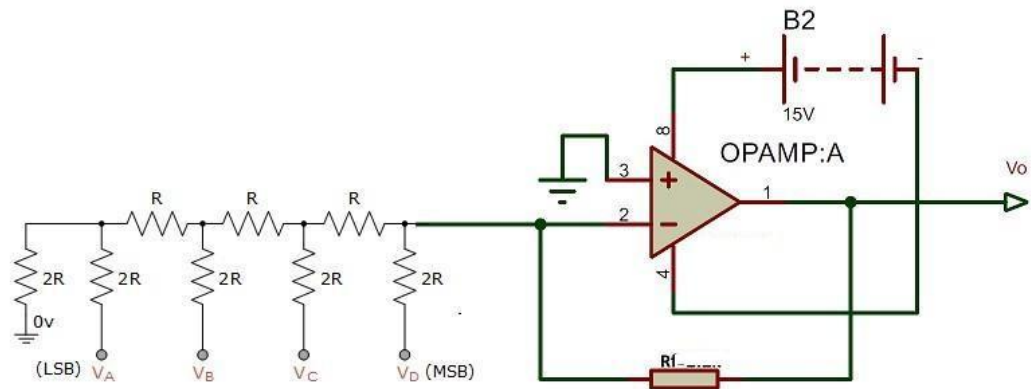
The following components are required to perform this experiment.

No.	Components	Specifications / Values	Quantity
1	Resistors	1k	Use these resistors in combination.
2	Op-Amp	LM741	1

Table 3: Components required to perform In-Lab Task 1.

Procedure

Implement the following circuit shown in Figure



1. Using an equation (2), calculate the theoretical values R2R DAC
2. Obtain all the experimental values of R-2R ladder network
3. Measure voltage level using DMM and record
4. Now calculate percentage error in experimental value by following formula,

$$\text{Percentage Error} = \frac{\text{theoretical value} - \text{experimental value}}{\text{theoretical value}} \times 100$$

S.No	Binary	R2R theory	R2R experiment	Error (%)
0	0 0 0 0	0.0000	0.0002	N/A
1	0 0 0 1	-0.3125	-0.298	4.64
2	0 0 1 0	-0.6250	-0.598	4.32
3	0 0 1 1	-0.9375	-0.911	2.83
4	0 1 0 0	-1.2500	-1.2	4.00
5	0 1 0 1	-1.5625	-1.55	0.80
6	0 1 1 0	-1.8750	-1.788	4.64
7	0 1 1 1	-2.1875	-2.1	4.00
8	1 0 0 0	-2.5000	-2.53	1.20
9	1 0 0 1	-2.8125	-2.764	1.72
10	1 0 1 0	-3.1250	-3.06	2.08
11	1 0 1 1	-3.4375	-3.24	5.75
12	1 1 0 0	-3.7500	-3.69	1.60
13	1 1 0 1	-4.0625	-4.004	1.44

14	1 1 1 0	-4.3750	-4.31	1.49
15	1 1 1 1	-4.6875	-4.61	1.65

Conclusion:

Compare both digital to analog converters in terms of design and performance.

Both circuits successfully convert a 4-bit digital input into a proportional analog output voltage, confirming the operating principle of Eq. 1 and Eq. 2.

In terms of design, the R-2R ladder is simpler and more practical to build and scale, since it needs only two resistor values (R and 2R) for any number of bits, while the binary weighted resistor DAC needs a distinct, precisely matched resistor for every bit, which is hard to manufacture accurately as resolution increases.

In terms of performance, both circuits tracked their theoretical values closely: the BWR errors were mostly under 1.5% and the R-2R errors were mostly under 5%, with small deviations caused by resistor tolerances, breadboard contact resistance, and op-amp offset.

Overall, although the BWR DAC gave a slightly lower percentage error in this experiment, the R-2R ladder is the more scalable and commercially practical design for higher-resolution DACs.

What is the voltage step size in task 1 and task 2

Task 1 (Binary Weighted Resistor DAC): step size = $V_{\text{ref}}/16 = 5 \text{ V} / 16 = 0.625 \text{ V}$, which matches the 0.625 V spacing between consecutive theoretical values in Table 2.

Task 2 (R-2R Ladder DAC): step size = $V_{\text{ref}}/16 = 5 \text{ V} / 16 = 0.3125 \text{ V}$, which matches the 0.3125 V spacing between consecutive theoretical values in Table 4.

Both step sizes equal V_{ref} divided by 2^n ($n = 4$ bits); the BWR step is larger because its feedback/input resistor ratio differs from the R-2R network's.

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Lab#04 Digital to Analog Conversion

Lab #4 Marks distribution:

		LR1=30	LR3=20	LR5=20	LR6=10	AR4=20
In-Lab Tasks	Task 1	15	10	10	10	20
	Task 2	15	10	10		
Total Marks						

Lab Evaluation Assessment Rubric

#	Assessment Elements	(0<Level 1<=4)	(4< Level 2<=6)	(6< Level 3<=8)	(8< Level 4<=10)
LR1	Neat and Clean Circuit layout	Components are wired and didn't show neat and clean connections and minimal efforts shown.	Components are wired with untidy connection and didn't show neat and clean connections	Few but not all Components are wired with neat and clean connections	Complete components are wired with neat and clean tight connections. Given task completed in due time.
LR3	Troubleshooting	Unable to identify the fault/minimal effort shown	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and able to make necessary steps and actions to correct it.
LR4	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included	Measurements are somewhat inaccurate and very imprecise. Observations are incomplete or recorded in a confusing way. Major errors using symbols, units and significant digits	Measurements are mostly accurate. Observations are generally complete. Minor errors using symbols, units and significant digits	Measurements are both accurate and precise. Observations are very thorough. Includes appropriate symbols, units and significant digits and completed in due time.
LR5	Results & Plots	Figures, graphs, tables contain errors or are poorly constructed, have missing titles, captions, units missing or incorrect, etc.	Most figures, graphs, tables OK, some still missing some important or required features	All figures, graphs, tables are correctly drawn, but some have minor problems or could still be improved	All figures, graphs, tables are correctly drawn and contain titles/captions.
LR6	Clean-up	All equipment/PC is not powered off. All items left at station and not cleaned.	Many equipment/PC is not powered off. Many items left at station.	Some equipment/PC is not powered off. Some items left at station.	All equipment/PC is powered off. Station left neat and clean.
LR8	Contribution	No participation towards the group tasks	Slight participation towards the group tasks	Substantial participation towards the group tasks	Outstanding participation towards the group tasks
LR9	Report	No summary provided. The number/ amount of tasks completed below the level of satisfaction and/or submitted late.	Couldn't provide good summary of in-lab tasks. Some major tasks were completed but not explained well. Submission on time. Some major plots and figures provided.	Good summary of In-lab tasks. All major tasks completed except few minor ones. The work is supported by some decent explanations, Submission on time, All necessary plots, and figures. provided	Outstanding Summary of In-Lab tasks, All task completed and explained well, submitted on time, good presentation of plots and figure with proper label, titles and captions.
AR4	*Report Submission	Late submission after 1 week.	Late submission after 4 days and within a week.	Late submission after the lab timing and after 2 days of the due date.	Timely submission of the report and in lab time.

***Report:** Report will not be accepted after 1 week of due date